



National Radio  
Astronomy  
Observatory

NAASC  
NORTH AMERICAN ALMA SCIENCE CENTER

## ALMA Proposal Preparation Workshop

March 25, 2026 @INCT-UDA, Copiapó

# 1.1. Basics of Radio Interferometry

Carla Cornil-Baiotto

[carla.cornil@postgrado.uv.cl](mailto:carla.cornil@postgrado.uv.cl)



## ➤ Radio regime

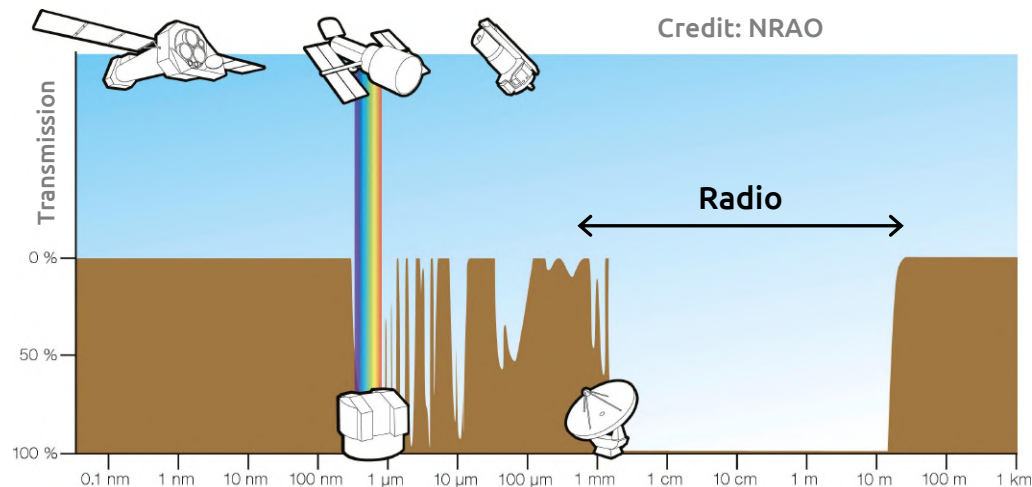
- Submm to meter wavelengths
- Cold, dusty environments
- Complementary to optical/IR
- Jansky:  $1 \text{ Jy} = 10^{-26} \text{ W m}^{-2} \text{ Hz}^{-1}$

## ➤ Physical processes

- Thermal emission from dust and ionized gas
- Spectral lines from atomic and molecular transitions
- Non-thermal processes (synchrotron, free-free cont.)

## ➤ Angular reso: $\theta_{\text{res}} \propto \lambda/D$

- Long  $\lambda$  implies poor  $\theta_{\text{res}}$  (1 dish)
- Large apertures are required



IRAM (30m)



GBT (100m)

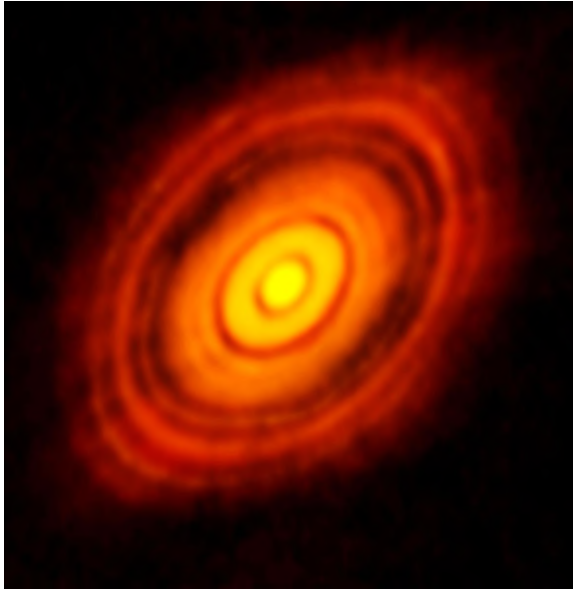


FAST (500m)

## Origins of planets

HL Tau protoplanetary disk

ALMA Partnership et al. (2015)

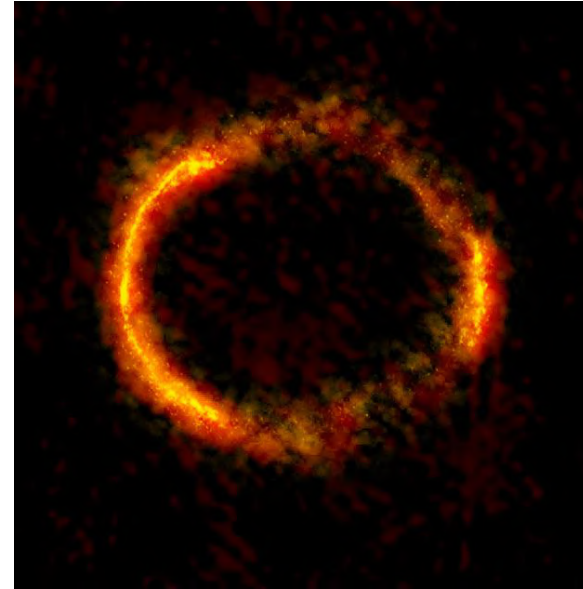


1.4 arcsec

## Galaxy evolution

Lensed galaxy SDP.81 at  $z = 3$

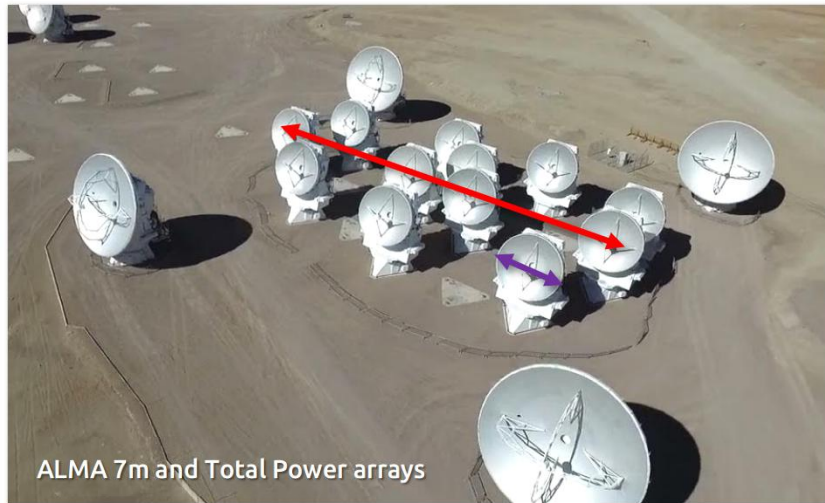
ALMA Partnership et al. (2015)



4 arcsec

- Some science cases require angular resolutions well beyond what single-dish telescopes can provide → use of radio interferometry

- The FOV becomes the diffraction limit of a **single antenna**.
- Resolution of a telescope equivalent in size to the **largest baseline**.



## Angular Resolution

$$\theta_{\text{res}} \sim \lambda / B_{\text{max}}$$

Largest distance between antennas

## FOV or Primary Beam

$$\theta_{\text{PB}} \sim \lambda / D$$

Diffraction limit of a single antenna

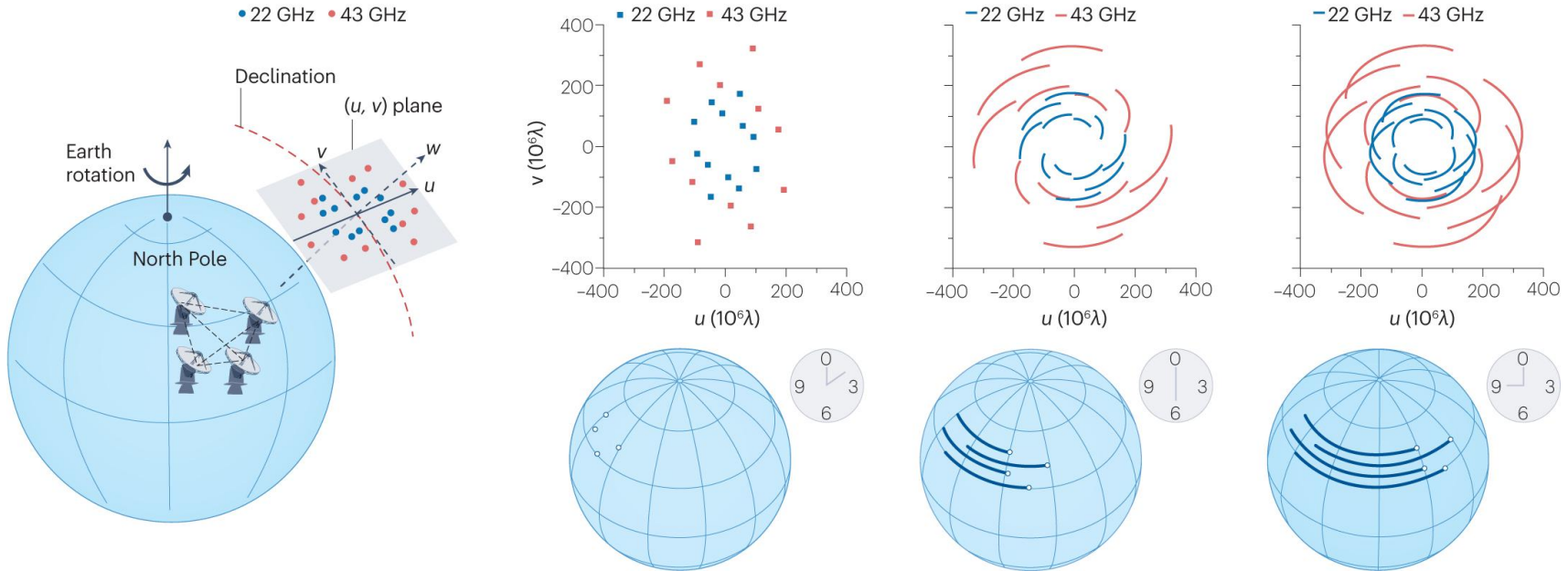
## Maximum Recoverable Scale

$$\theta_{\text{MRS}} \sim \lambda / B_{\text{min}}$$

Shortest distance between antennas

- Aperture synthesis: separate antennas working as a single telescope
- Data taken in the uv-plane: **Amplitude** (how much) & **Phase** (where)

Aperture Synthesis  
Physics Nobel Prize 1974  
M. Ryle & A. Hewish



Asaki et al. (2023), *Nature Reviews*

Each pair of antennas samples a point in the  $uv$ -plane, and the sky is scanned through Earth rotation and antenna motion.

- Complex visibility in the Fourier domain:

$$V(u, v) = \iint T(l, m) e^{-2\pi i (ul + vm)} dl dm$$

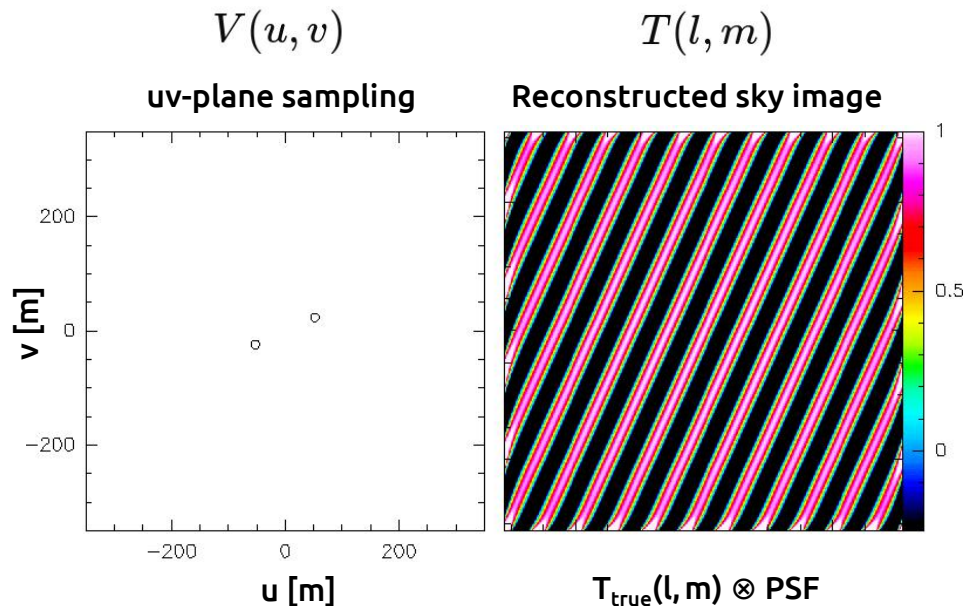
- Sky brightness in the Image domain:

$$T(l, m) = \iint V(u, v) e^{+2\pi i (ul + vm)} du dv$$

- Van Cittert–Zernike theorem:

$$V(u, v) = \text{FT}\{ T(l, m) \}$$

$$T(l, m) = \text{FT}^{-1}\{ V(u, v) \}$$



Credit: NRAO

- Complex visibility in the Fourier domain:

$$V(u, v) = \iint T(l, m) e^{-2\pi i (ul + vm)} dl dm$$

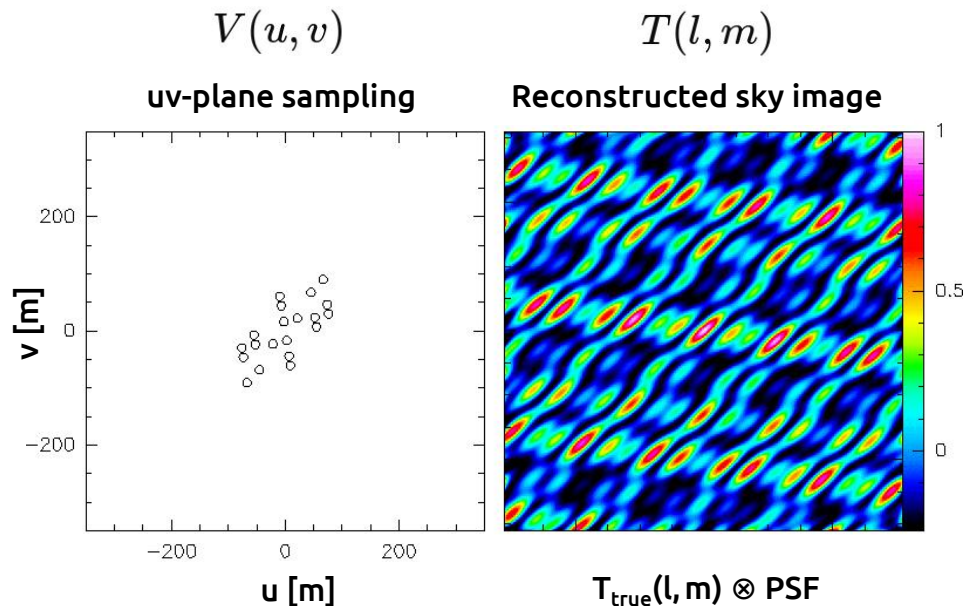
- Sky brightness in the Image domain:

$$T(l, m) = \iint V(u, v) e^{+2\pi i (ul + vm)} du dv$$

- Van Cittert-Zernike theorem:

$$V(u, v) = \text{FT}\{ T(l, m) \}$$

$$T(l, m) = \text{FT}^{-1}\{ V(u, v) \}$$



Credit: NRAO

- Complex visibility in the Fourier domain:

$$V(u, v) = \iint T(l, m) e^{-2\pi i (ul + vm)} dl dm$$

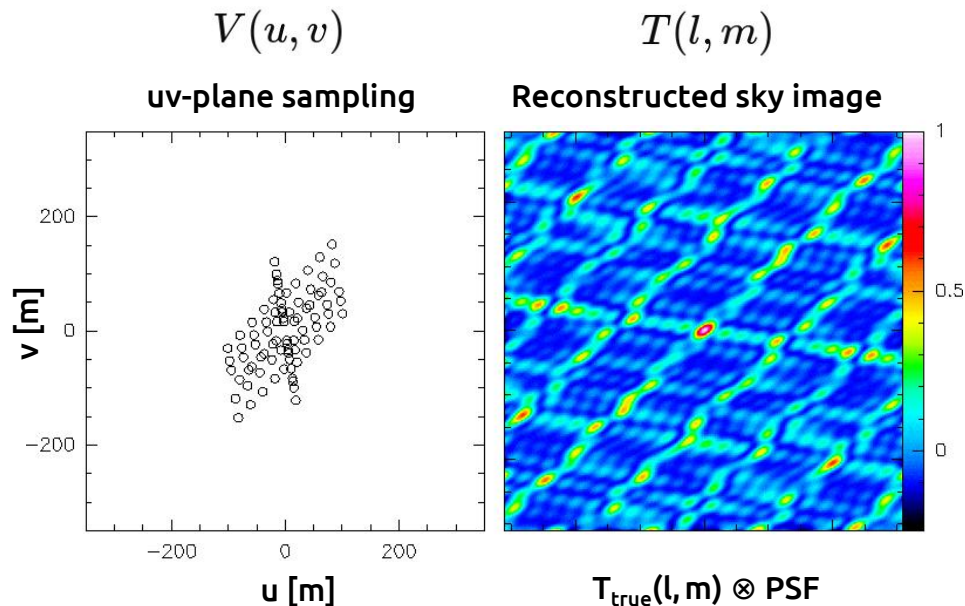
- Sky brightness in the Image domain:

$$T(l, m) = \iint V(u, v) e^{+2\pi i (ul + vm)} du dv$$

- Van Cittert–Zernike theorem:

$$V(u, v) = \text{FT}\{ T(l, m) \}$$

$$T(l, m) = \text{FT}^{-1}\{ V(u, v) \}$$



Credit: NRAO

- Complex visibility in the Fourier domain:

$$V(u, v) = \iint T(l, m) e^{-2\pi i (ul + vm)} dl dm$$

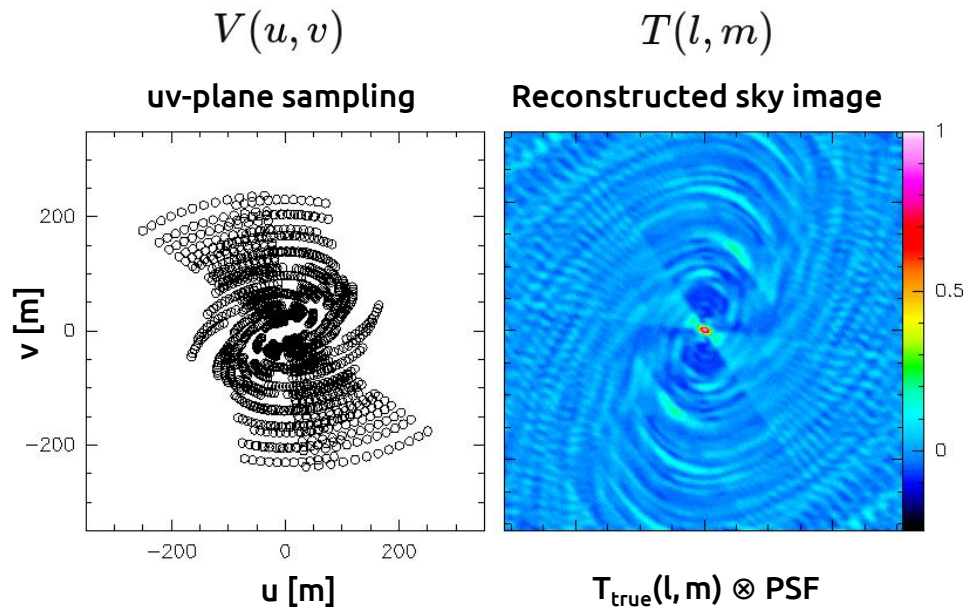
- Sky brightness in the Image domain:

$$T(l, m) = \iint V(u, v) e^{+2\pi i (ul + vm)} du dv$$

- Van Cittert–Zernike theorem:

$$V(u, v) = \text{FT}\{T(l, m)\}$$

$$T(l, m) = \text{FT}^{-1}\{V(u, v)\}$$



Credit: NRAO

- Complex visibility in the Fourier domain:

$$V(u, v) = \iint T(l, m) e^{-2\pi i (ul + vm)} dl dm$$

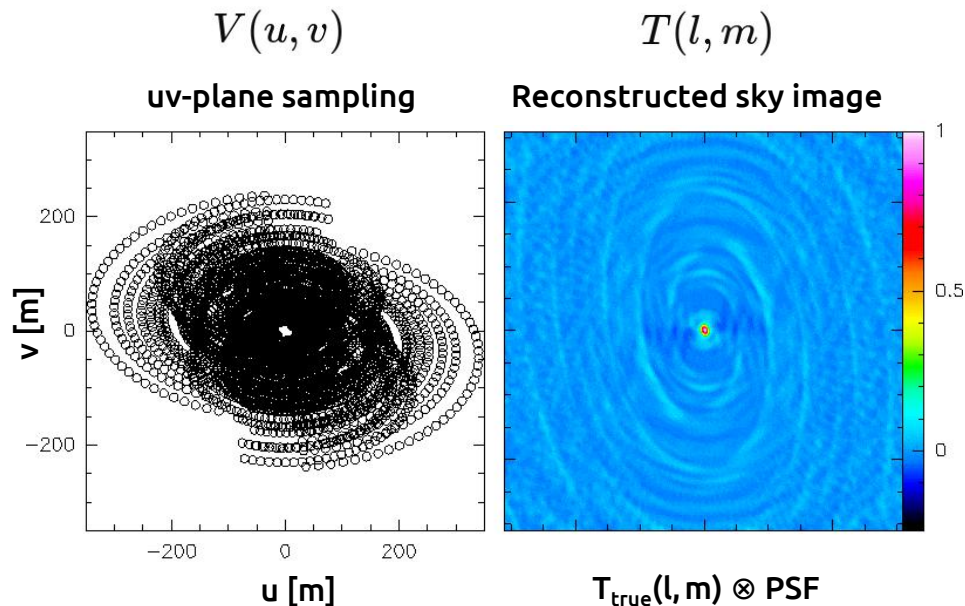
- Sky brightness in the Image domain:

$$T(l, m) = \iint V(u, v) e^{+2\pi i (ul + vm)} du dv$$

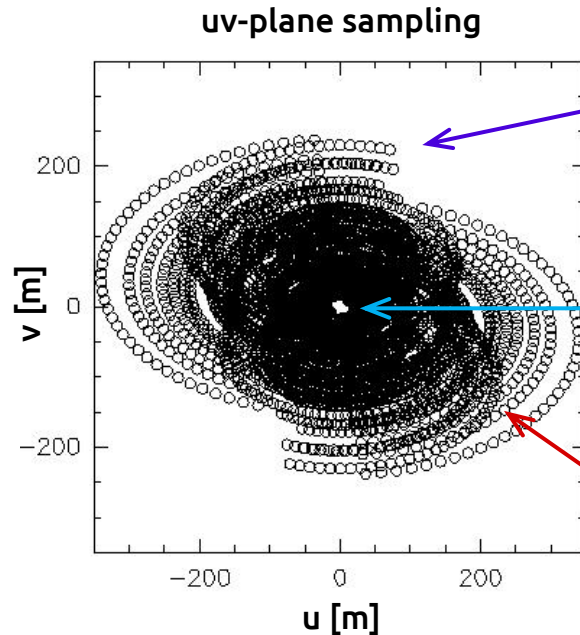
- Van Cittert–Zernike theorem:

$$V(u, v) = \text{FT}\{T(l, m)\}$$

$$T(l, m) = \text{FT}^{-1}\{V(u, v)\}$$



Credit: NRAO



## Outer boundary

Sets the angular resolution limit:  $\theta_{\text{res}} = \lambda/B_{\text{max}}$

## Inner hole

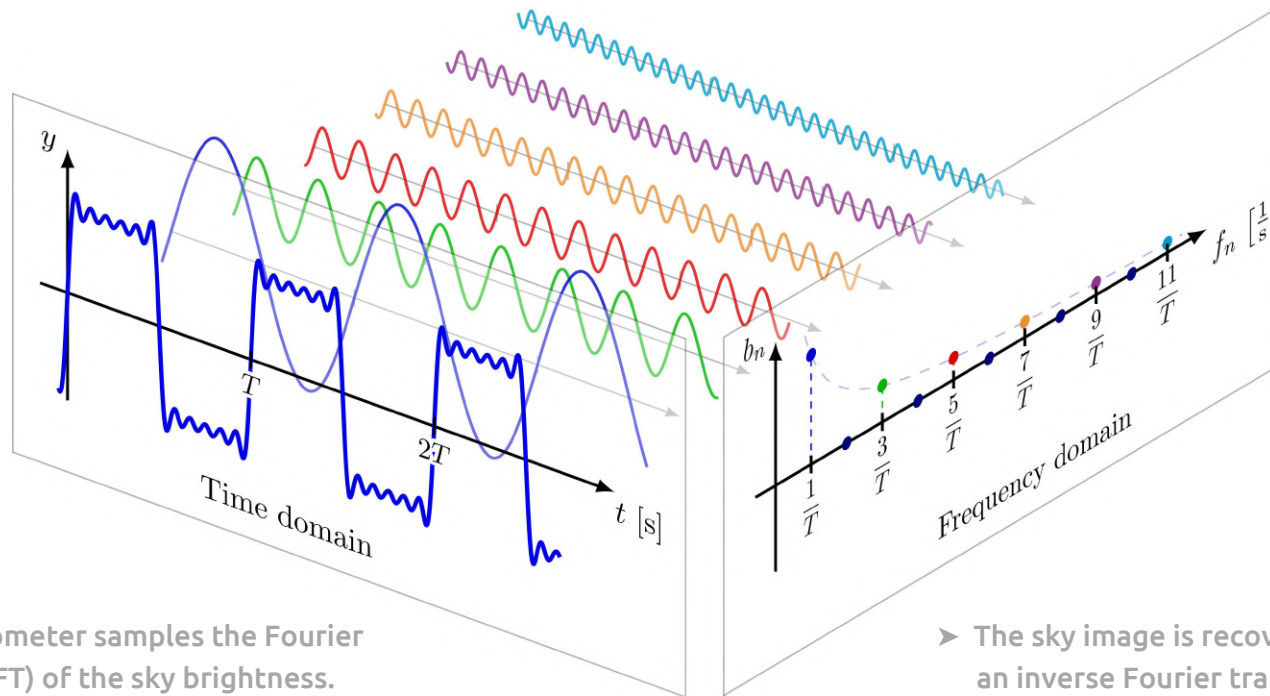
- Insensitive to large angular scales ( $\theta > \lambda/B_{\text{min}}$ )
- Sets the maximum "recoverable scale"

## Incomplete sampling

- Missing info: imperfect image reconstruction
- Creates sidelobes in the synthesized beam

# Relation between Visibilities and Sky Brightness

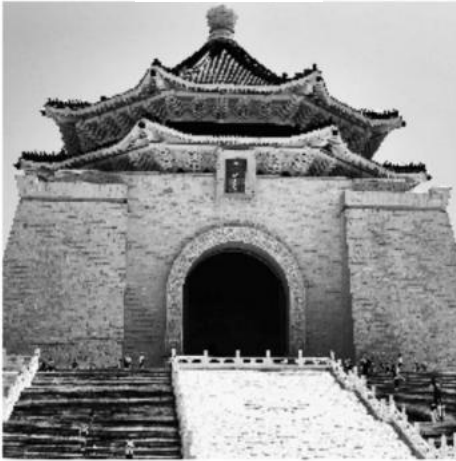
- Sky brightness distribution can be represented as a sum of pure frequencies
- Interferometers measure these Fourier components ( $\equiv$  visibilities)



- An interferometer samples the Fourier transform (FT) of the sky brightness.

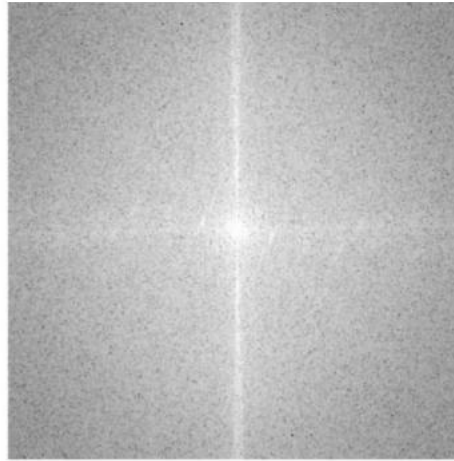
- The sky image is recovered via an inverse Fourier transform.

$I(l, m)$

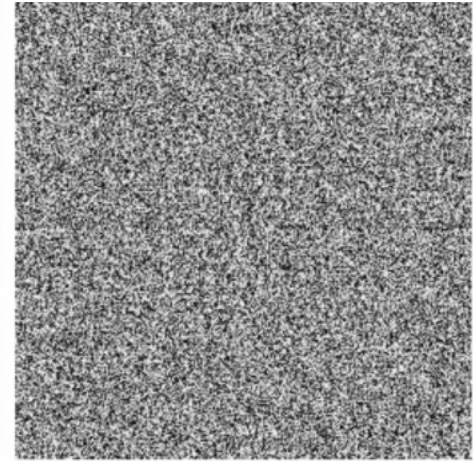


$\mathcal{F}$   
 $\Rightarrow$

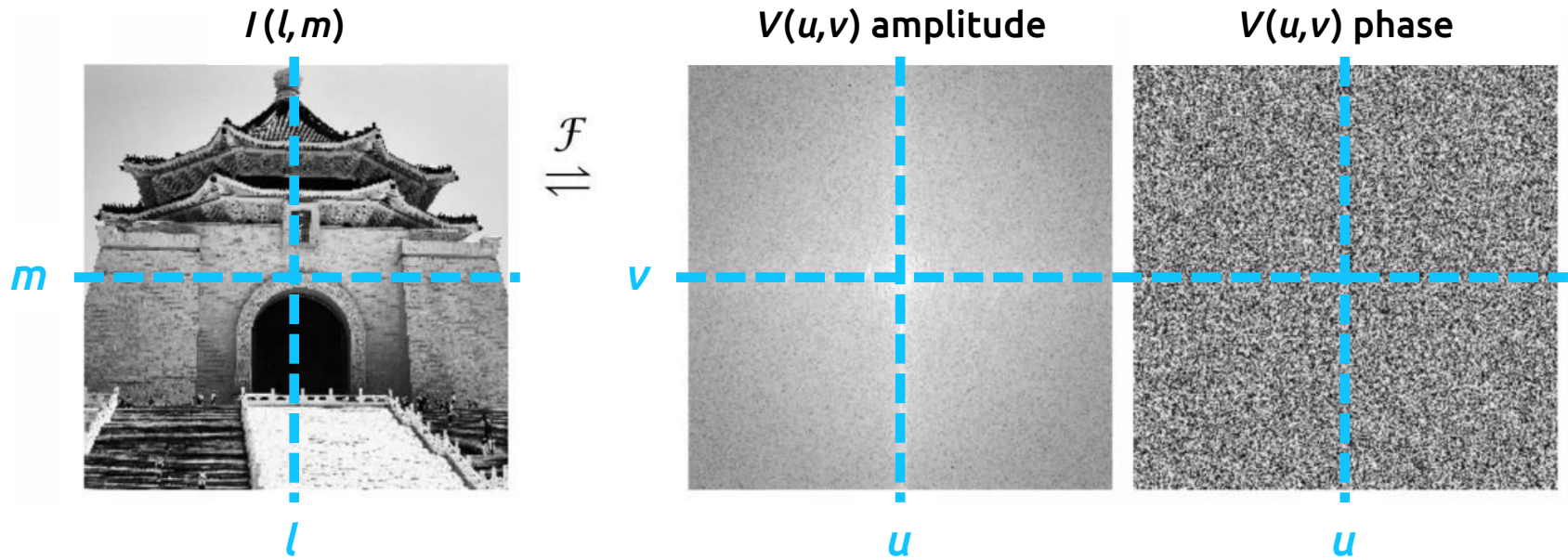
$V(u, v)$  amplitude



$V(u, v)$  phase



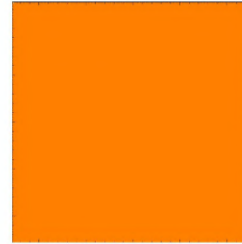
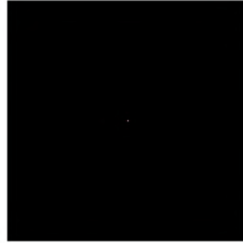
# Relation between Visibilities and Sky Brightness



## Examples of Fourier Transform Pairs

Sky brightness  $I(l, m)$

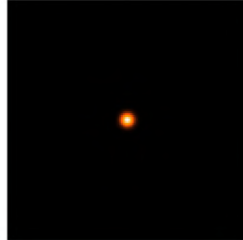
$\delta$  Function



$\text{Amp}\{ V(u, v) \}$

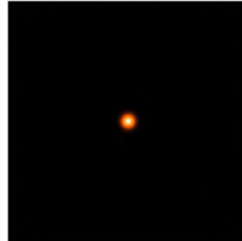
Constant

Gaussian



Gaussian

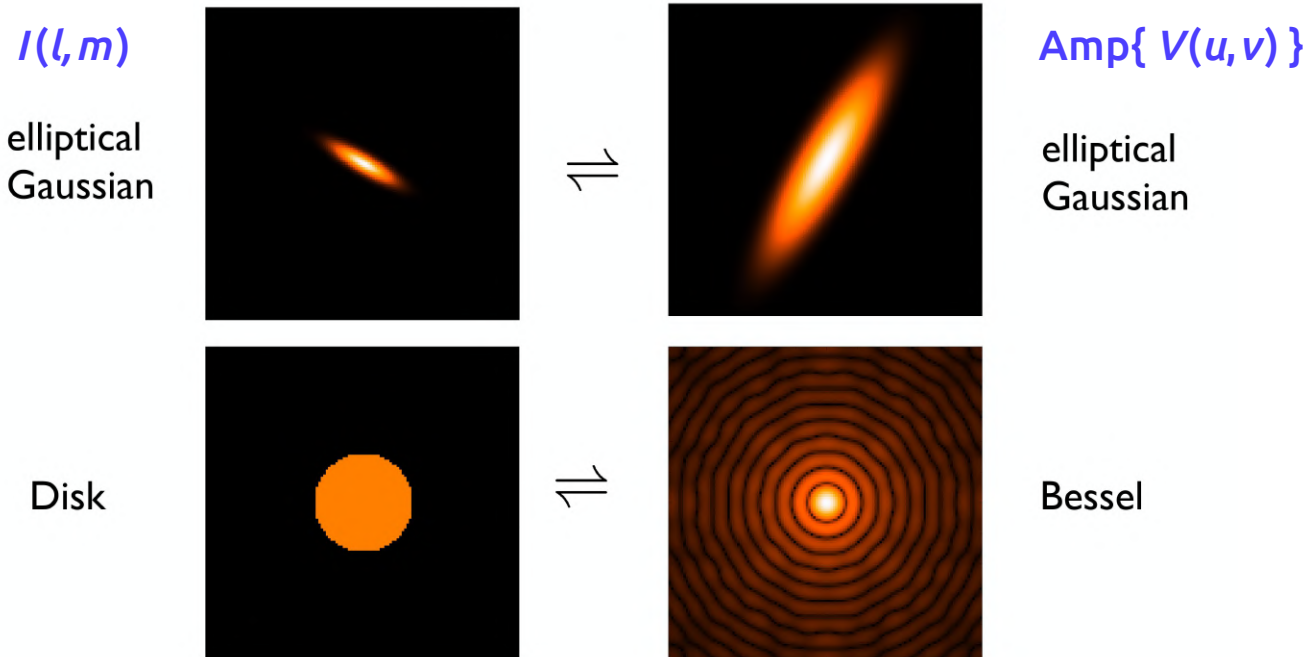
Gaussian



Gaussian

➤ Narrow features transform to wide features (and vice-versa).

## Examples of Fourier Transform Pairs

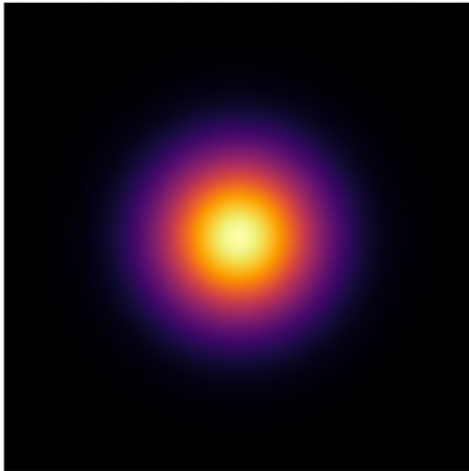


- Sharp edges produce many high spatial frequencies, leading to sinc-like ringing patterns (Gibbs phenomenon).

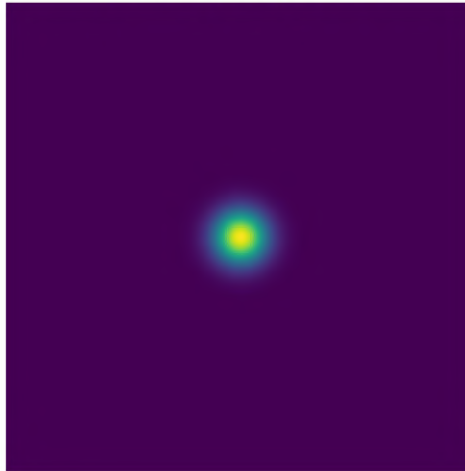
# Amplitude and Phase

- Amplitude: **how much** of a frequency component is present.
- Phase: **where** this frequency component is located.

Sky brightness:  $I(l,m)$



Visibility amplitude



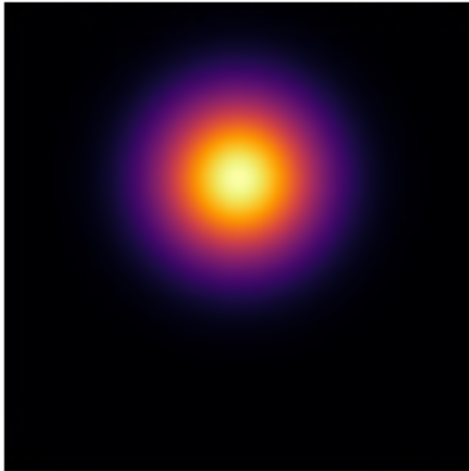
Visibility phase



# Amplitude and Phase

- Amplitude: **how much** of a frequency component is present.
- Phase: **where** this frequency component is located.

Sky brightness:  $I(l,m)$



Visibility amplitude



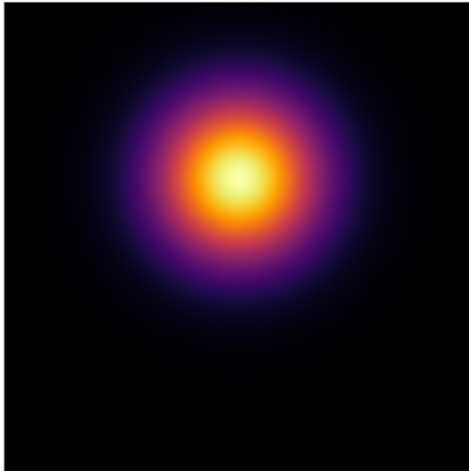
Visibility phase



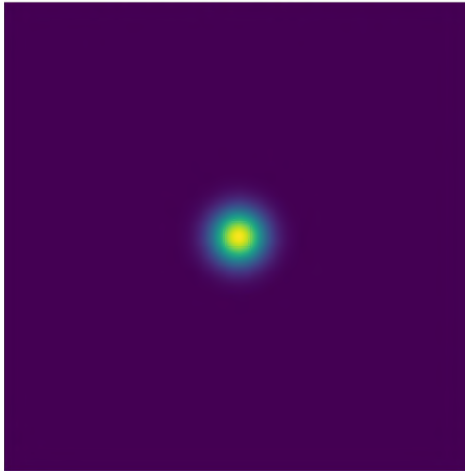
# Amplitude and Phase

- Amplitude: **how much** of a frequency component is present.
- Phase: **where** this frequency component is located.

Sky brightness:  $I(l,m)$



Visibility amplitude



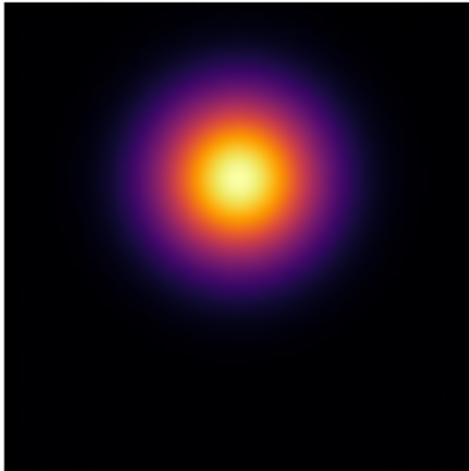
Visibility phase



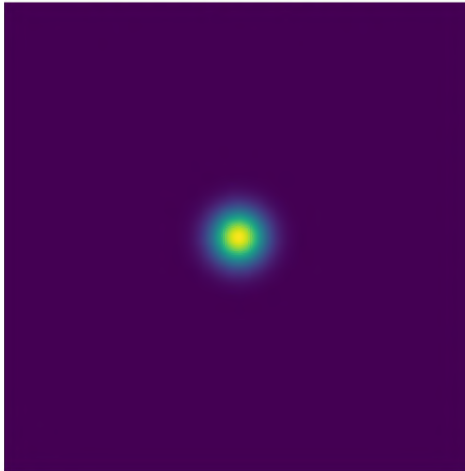
# Amplitude and Phase

- Amplitude: **how much** of a frequency component is present.
- Phase: **where** this frequency component is located.

Sky brightness:  $I(l,m)$



Visibility amplitude



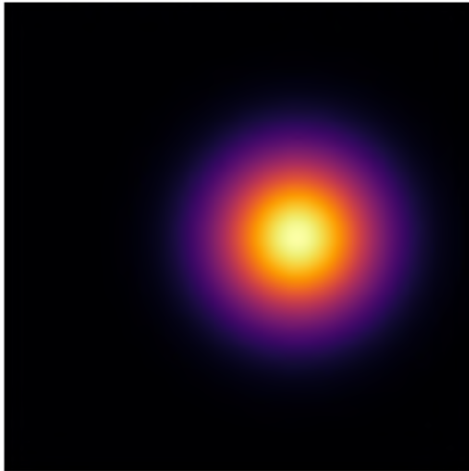
Visibility phase



# Amplitude and Phase

- Amplitude: **how much** of a frequency component is present.
- Phase: **where** this frequency component is located.

Sky brightness:  $I(l,m)$



Visibility amplitude



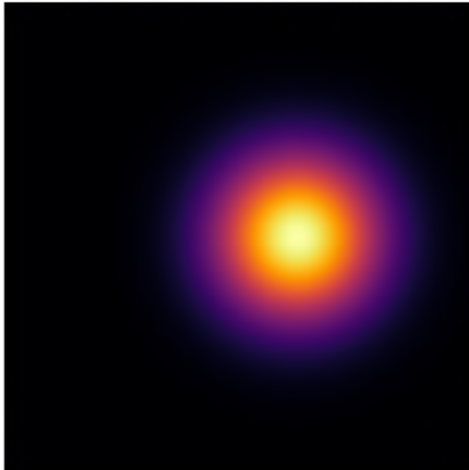
Visibility phase



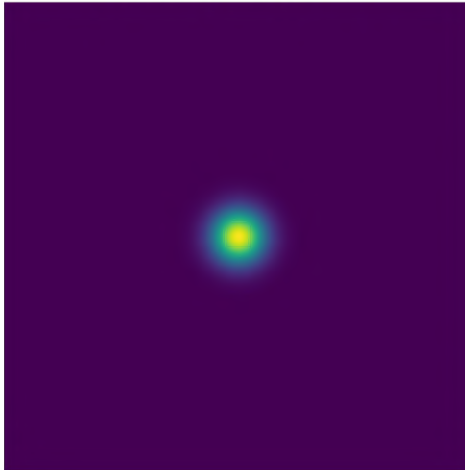
# Amplitude and Phase

- Amplitude: **how much** of a frequency component is present.
- Phase: **where** this frequency component is located.

Sky brightness:  $I(l,m)$



Visibility amplitude



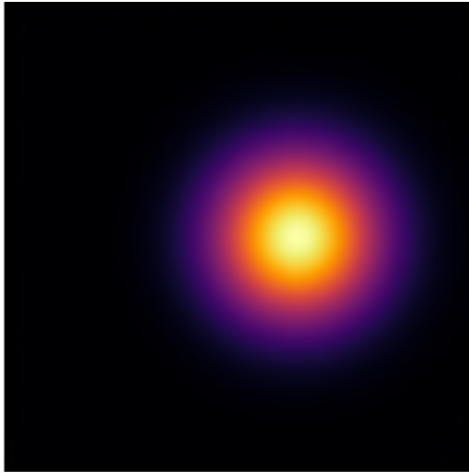
Visibility phase



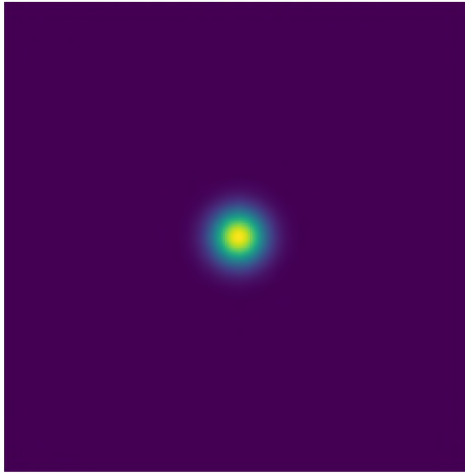
# Amplitude and Phase

- Amplitude: **how much** of a frequency component is present.
- Phase: **where** this frequency component is located.

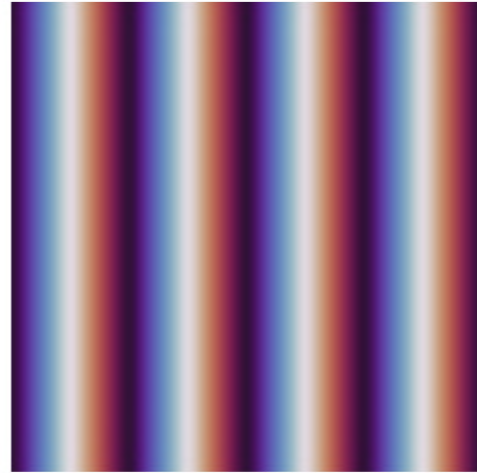
Sky brightness:  $I(l,m)$



Visibility amplitude



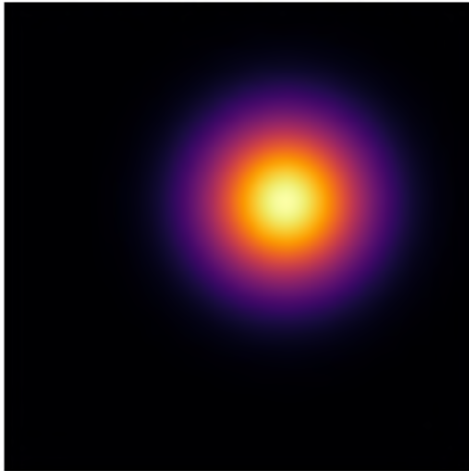
Visibility phase



# Amplitude and Phase

- Amplitude: **how much** of a frequency component is present.
- Phase: **where** this frequency component is located.

Sky brightness:  $I(l,m)$



Visibility amplitude



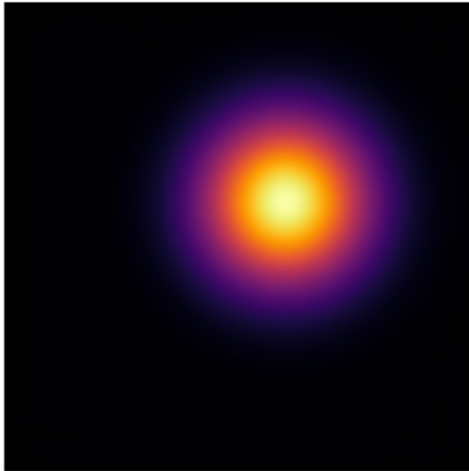
Visibility phase



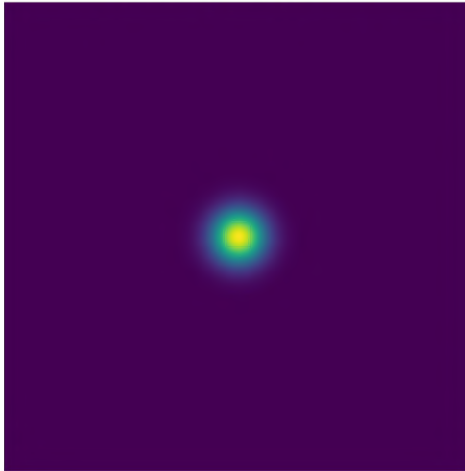
# Amplitude and Phase

- Amplitude: **how much** of a frequency component is present.
- Phase: **where** this frequency component is located.

Sky brightness:  $I(l,m)$



Visibility amplitude

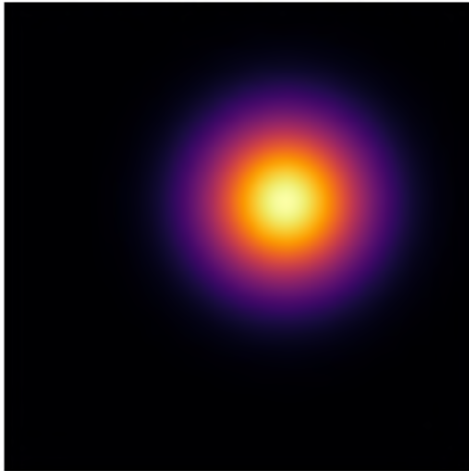


Visibility phase

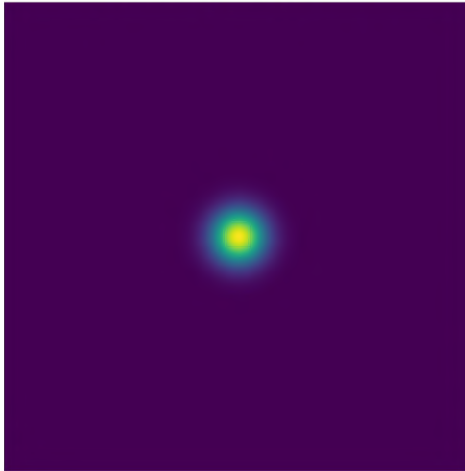


- Amplitude: **how much** of a frequency component is present.
- Phase: **where** this frequency component is located.

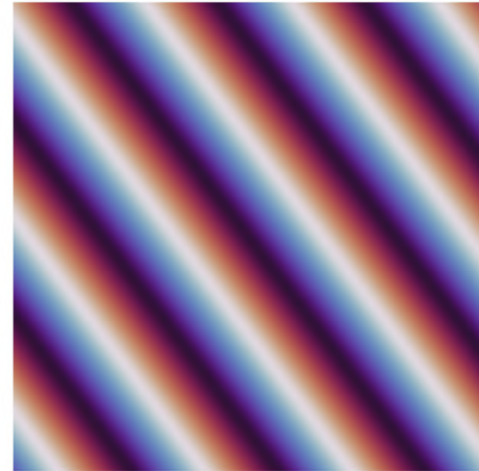
Sky brightness:  $I(l,m)$



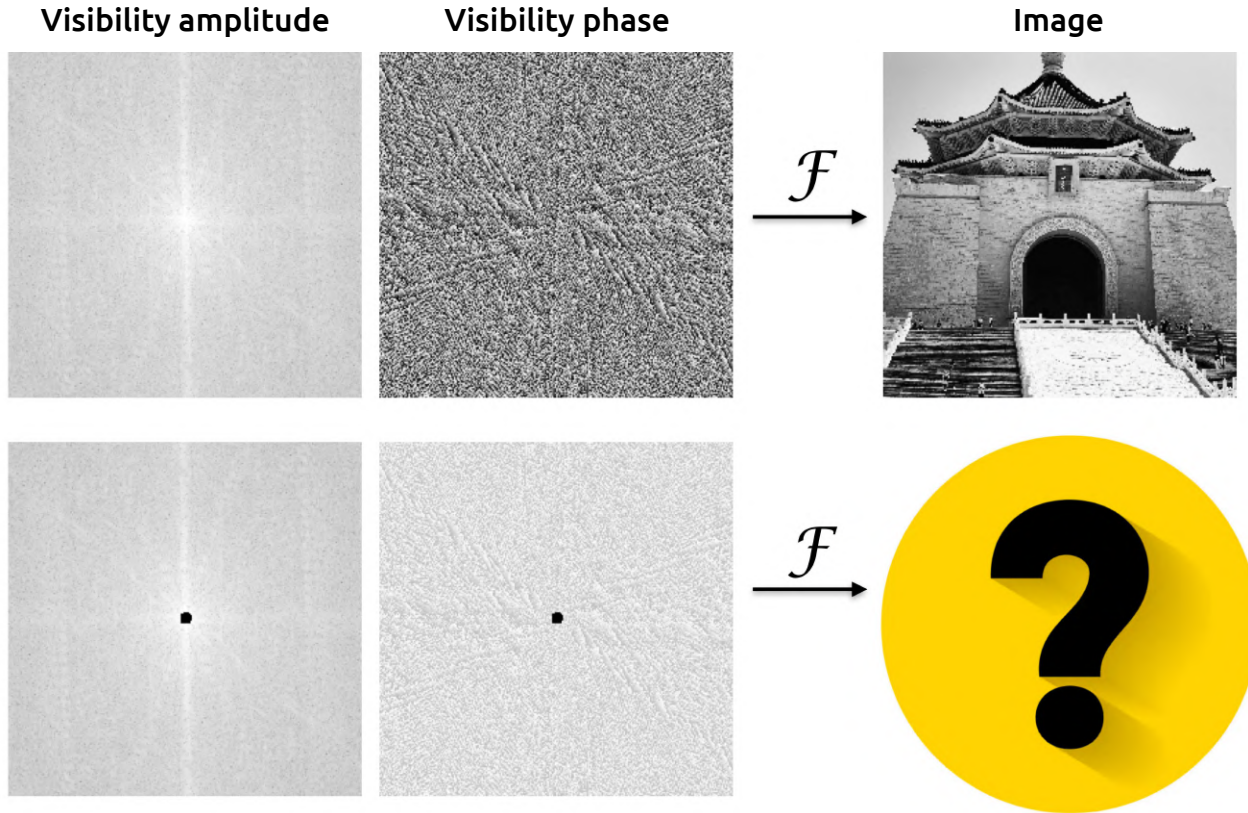
Visibility amplitude



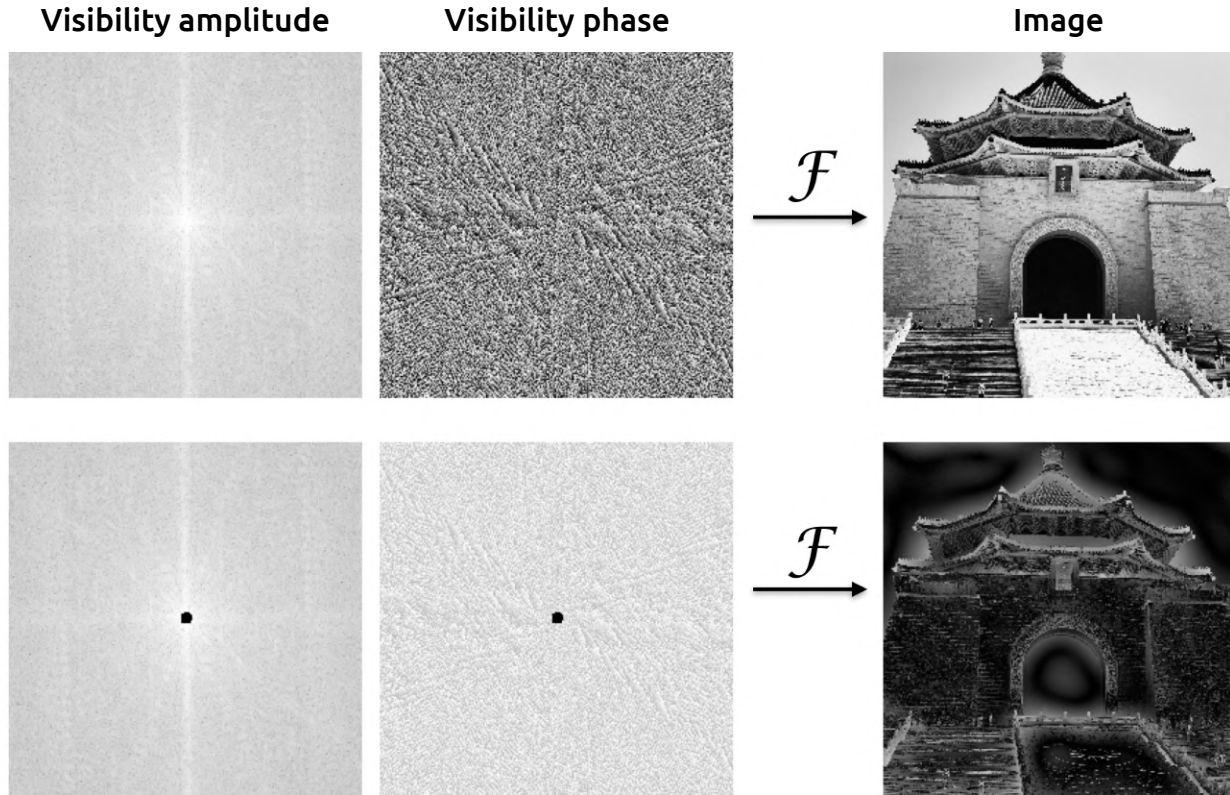
Visibility phase



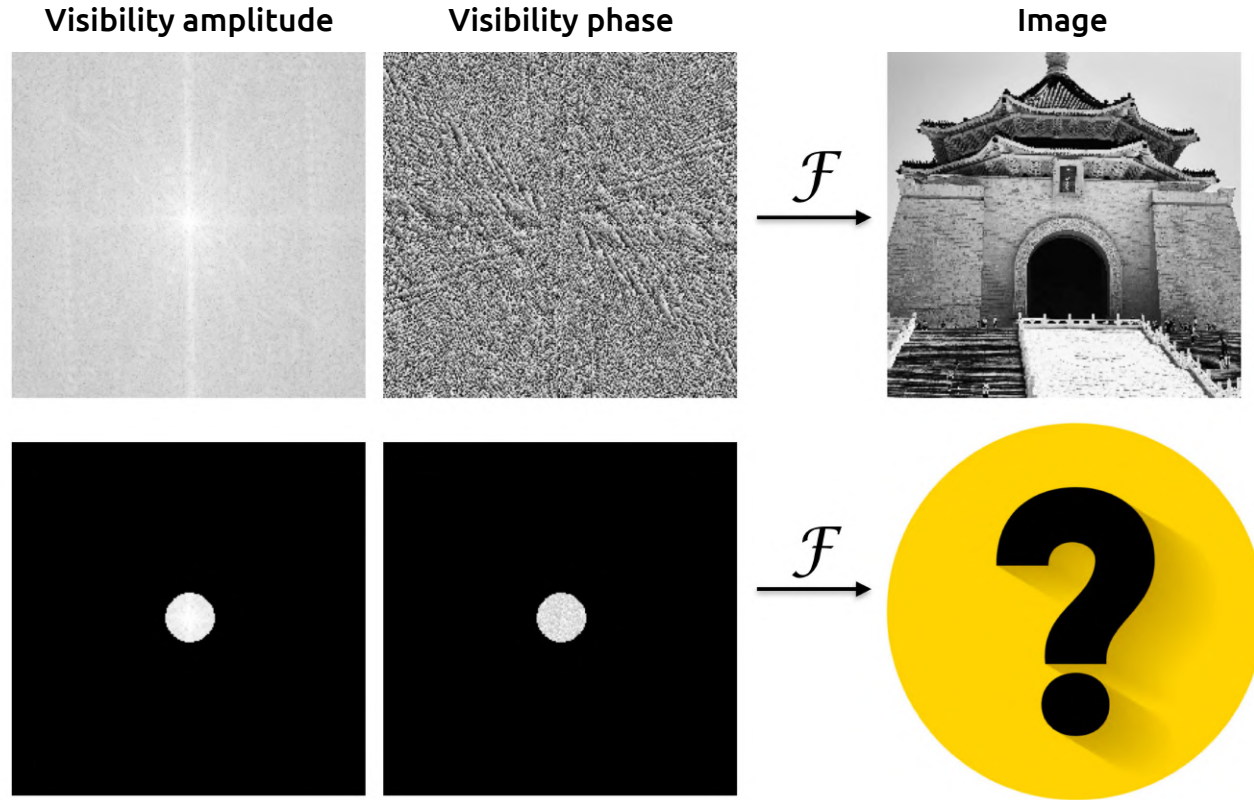
# Impact of Finite $uv$ -Sampling



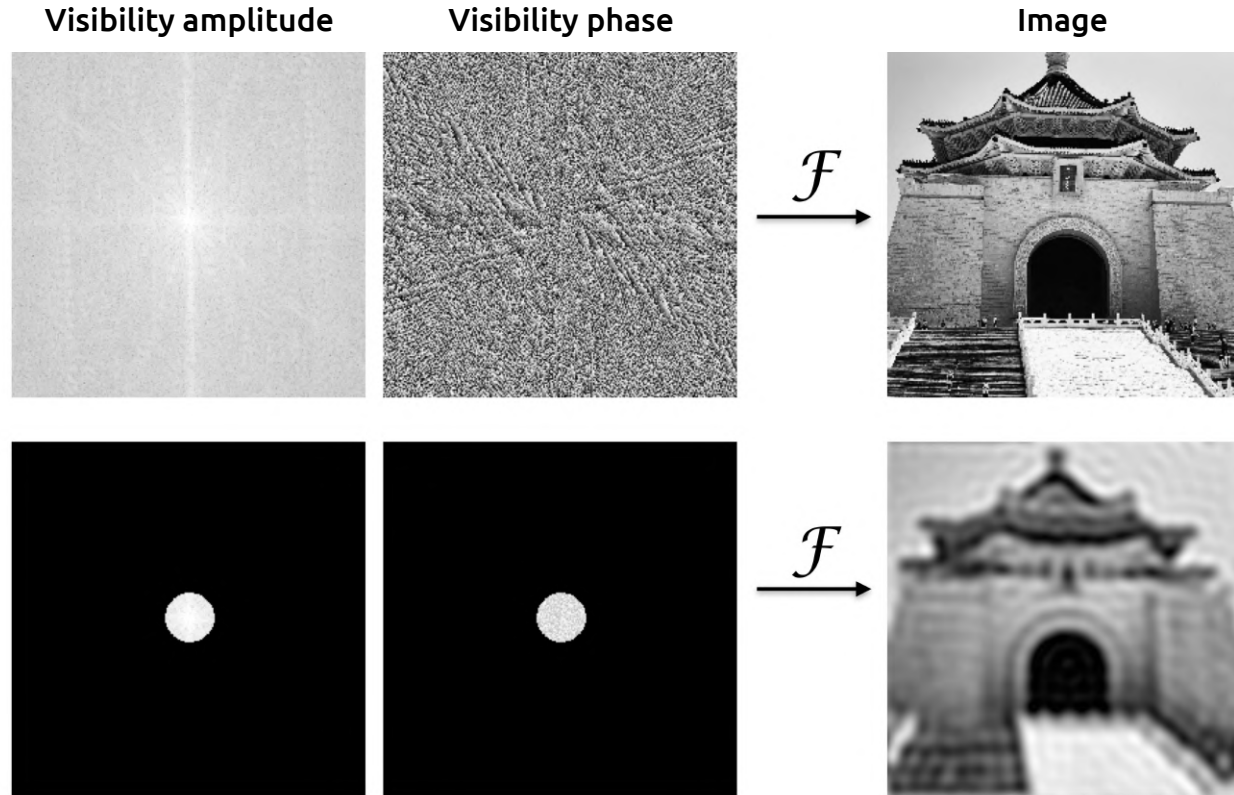
# Impact of Finite $uv$ -Sampling

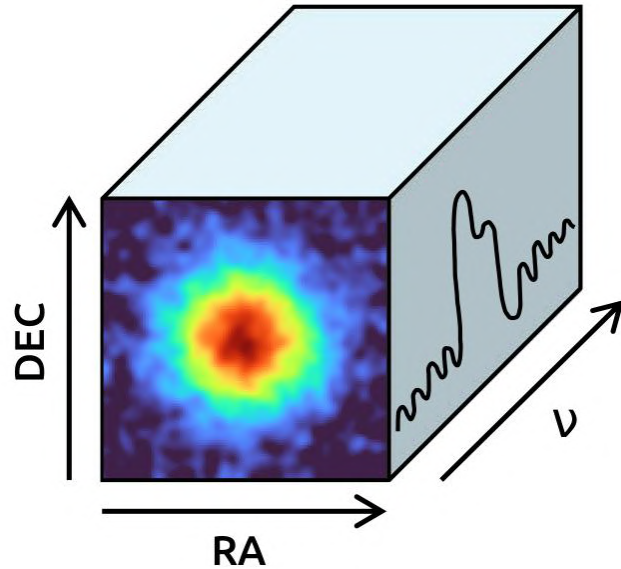


# Impact of Finite $uv$ -Sampling

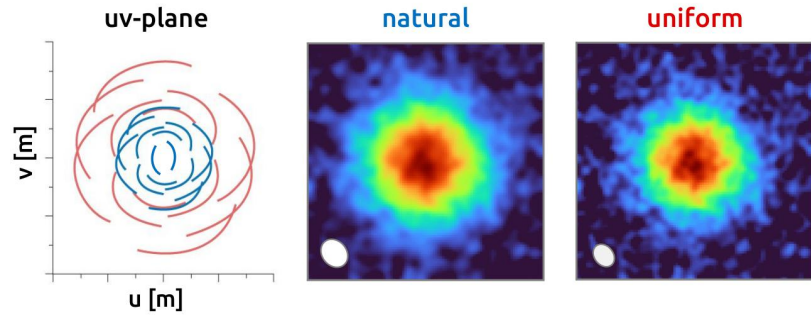


# Impact of Finite $uv$ -Sampling

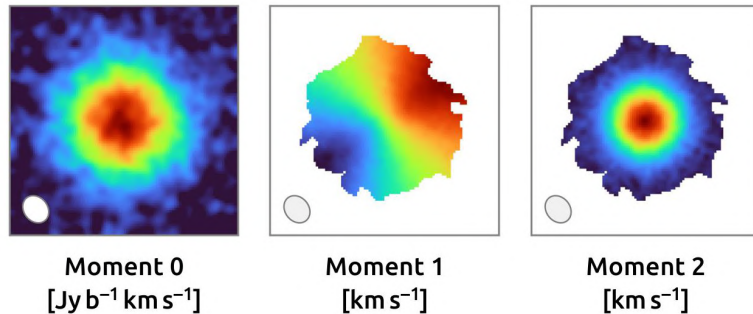


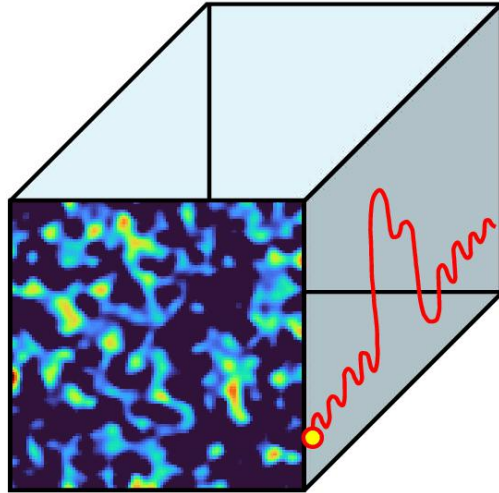


## ► Different data weighting schemes



## ► Spectral line moment maps

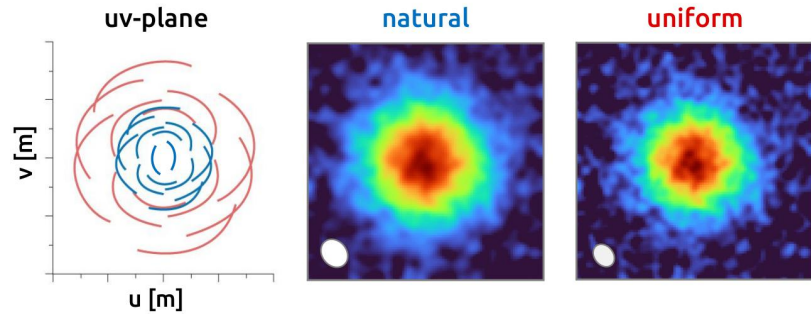




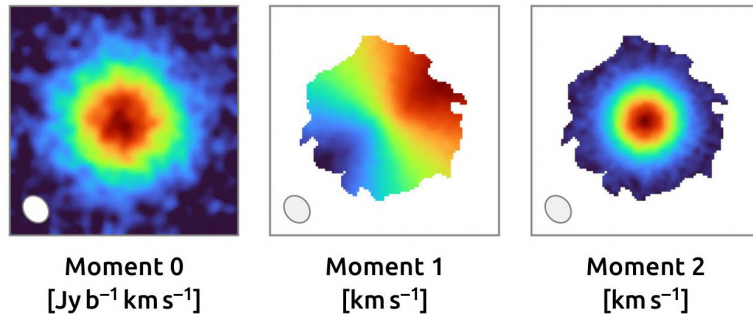
ALMA-related (animated) figures:

<https://drive.google.com/drive/folders/1JlXHTZqovMDQcEMiH5vDaevE4u5Wvzz0>

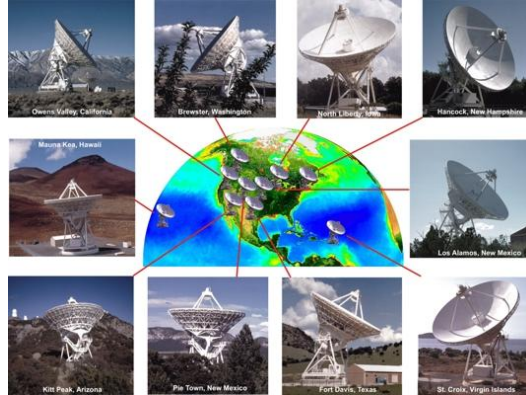
## ► Different data weighting schemes



## ► Spectral line moment maps



# Northern Hemisphere Interferometers



**VLBA (10 antennas), mainly USA**



**LOFAR (~50 stations)**  
**Netherlands**

LOFAR → meters (MHz)  
VLA → cm (GHz)  
VLBA → cm (GHz)  
NOEMA → (sub)mm  
(70–375 GHz)

# Southern Hemisphere Interferometers



ALMA → (sub)mm  
(35–950 GHz)

ATCA → cm/mm  
(1.1–105 GHz)

MeerKAT → cm  
(0.5–3.5 GHz)



MeerKAT (64 antennas)  
South Africa



ATCA (6 antennas)  
Australia

# Next-Generation Interferometers



## Square Kilometre Array Observatory

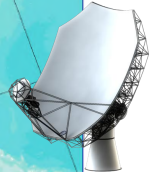
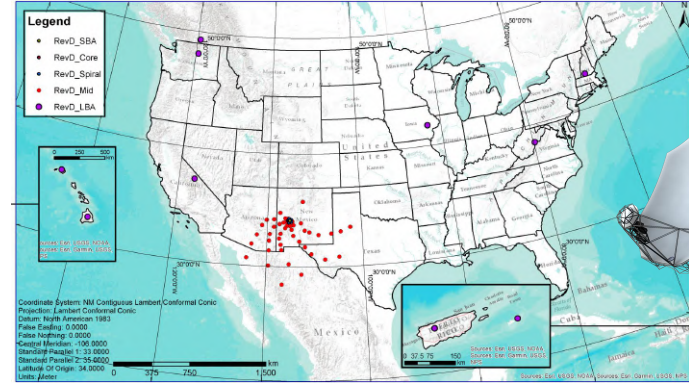


### Extreme sensitivity to faint, large-scale emission

- **SKA-Low** (Australia): 50 – 350 MHz (meters)
- **SKA-Mid** (South Africa): 0.35 – 15 GHz (cm)
- Largest collecting area ever built: **1 million m<sup>2</sup> of collecting area** to study critical scientific questions, from the birth of our Universe to the origins of life.



## Next Generation VLA (ngVLA)



### High-reso imaging bridging VLA and ALMA

- **1.2 – 116 GHz** Frequency Coverage
- **Array Design:** 244 x 18m offset Gregorian Antennas
  - **Core:** 114 fixed antennas;  $B_{\max} = 4.3$  km
  - **Spiral:** 54 fixed antennas;  $B_{\max} = 39$  km
  - **Mid:** 46 fixed antennas spread into NM, AZ, TX, MX;  $B_{\max} = 1070$  km
  - **Long:** 30 x 18m antennas located across continent;  $B_{\max} = 8860$  km
- **Short Baseline Array:** 19 x 6m offset Gregorian Antennas
  - Use 4 x 18m in **TP mode** to fill in ( $u, v$ ) hole.